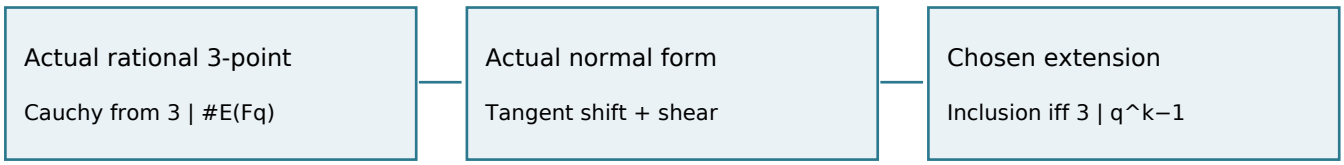
The qualified BK criterion at three, for all models

Result. Every nonsingular general Weierstrass model over every finite field, including characteristic two. With gcd(3,q)=1, 3 dividing the actual rational point-group cardinality and 3 not dividing q−1, all geometric points killed by three lie in a chosen compatible extension of cardinality q^k exactly when 3 divides q^k−1. Equivalently, that extension has exactly nine killed-three points, including the identity. No normalization, determinant, rank, pairing or abstract rationality premise. The qualified source survives this check at three. [pcc:unverified]



Actual AddEquiv + field-map naturality + all-n torsion + original nine-point bridge

Actual normalization. Additive Cauchy supplies an actual nonzero rational point P killed by three. Translate P to the origin and shear by its tangent slope. The curve equation proves a6=0, the slope proves a4=0 and the literal three-division polynomial proves a2=0. Actual nonsingularity proves the tangent denominator nonzero. The discriminant is preserved. No division by two is used. Coprimality supplies characteristic other than three through the additive order of 1. [pcc:unverified]

Actual group and descent bridges. The explicit coordinate map and inverse form an additive equivalence of actual Mathlib point groups. The proof handles identities, inverse pairs, secants and tangents. This equivalence commutes with actual field maps, compatible tower inclusions and point Frobenius. For every integer n it transports killed-point cardinalities and torsion containment in the chosen extension. Apply the preceding normal-form cubic-character criterion. The actual injective map on killed-three subgroups is surjective exactly when all geometric torsion descends; the preceding geometric nine-point theorem makes this equivalent to nine extension points. [pcc:unverified]

Skeptical controls. The credited actual F7 curve has 3^2 dividing its rational point count and q=1mod3 but a geometric three-torsion point not fixed by Frobenius. The genuine ordinary characteristic-three curve $y^2+xy+y=x^3$ has nonzero discriminant and only three geometric three-torsion points. Deleting coprimality refutes the unrestricted inclusion paraphrase there; the original nine-point condition is false too. These controls preserve the qualified theorem rather than refuting it. [pcc:unverified]

Primary and prior sources. BK1998 Theorem 1/proof/Remarks, pp142–143, DOI10.1007/s001459900040; Dujella section15.3, p494 coordinate/order-three normal form, discussed over Q. Pinned Mathlib variable-change/affine-point APIs, prior short scaling, actual F5/F7/F25 cases, general rank, Frobenius/descent and normal-form character proofs credited. Selected sections read; no full-book/paper read, novelty or corpus promotion. KB/Conductor unavailable; bounded fallback recorded. [pcc:unverified]

Validation and remaining scope. Whole gate 50 declarations, zero failures/trusted native exceptions. Complete statement binding and recursive custom-source/config pins. All 149 build-provenance files retained losslessly in 42 objects; five gate records retained. Pinned Lean4.35.0/Mathlib56a00585…, only propext/Classical.choice/Quot.sound. Exact local PCC replay and signed standing remain separate. Other primes ≥5, canonical Weil identification, full all-k p<500 census and remaining fourteen-item obligations are open. No native experiment, paid compute, upload, holes/custom axioms/native evaluator/raised limits/new global flags/waivers. [pcc:unverified]